

Finite Element Method

Approximations of Trial Solutions for 2D and 3D Problems — Part 3
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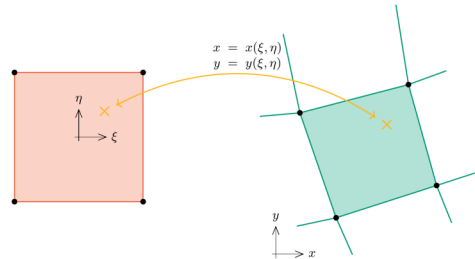
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- As seen in 1D FEM, integration of various forms of the shape functions over the domain of an element is required in formulating element matrices and vectors.
- We now show how the one-dimensional Gauss quadrature formulas developed in Section 4.6 are extended to two dimensions.
- Consider a typical integral defined over the physical domain of a quadrilateral element:

$$I = \int_{\Omega^e} f(\xi, \eta) d\Omega. \quad (1)$$

- Figure (right) shows the infinitesimal area $d\xi d\eta$ in the parent domain and its image in the physical domain.



Mapping from the parent domain (left) to the physical domain (right).

- The integral in Eq. (1) can be expressed as

$$I = \int_{-1}^1 \int_{-1}^1 |\mathbf{J}^e(\xi, \eta)| f(\xi, \eta) d\xi d\eta, \quad (2)$$

where the Jacobian matrix is given by

$$\mathbf{J}^e(\xi, \eta) = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \sum_I^{n_{en}} \frac{\partial N_I}{\partial \xi} x_I^e & \sum_I^{n_{en}} \frac{\partial N_I}{\partial \xi} y_I^e \\ \sum_I^{n_{en}} \frac{\partial N_I}{\partial \eta} x_I^e & \sum_I^{n_{en}} \frac{\partial N_I}{\partial \eta} y_I^e \end{bmatrix}. \quad (3)$$

- To evaluate the integral, we first carry out Gauss integration over ξ , which yields

$$I = \int_{-1}^1 \left(\int_{-1}^1 |\mathbf{J}^e(\xi, \eta)| f(\xi, \eta) d\xi \right) d\eta = \int_{-1}^1 \sum_{i=1}^{n_{gp}} W_i |\mathbf{J}^e(\xi_i, \eta)| f(\xi_i, \eta) d\eta. \quad (4)$$

- Next, integrating over η yields

$$I = \int_{-1}^1 \sum_{i=1}^{n_{gp}} W_i f(\xi_i, \eta) |\mathbf{J}^e(\xi_i, \eta)| d\eta = \sum_{i=1}^{n_{gp}} \sum_{j=1}^{n_{gp}} W_i W_j |\mathbf{J}^e(\xi_i, \eta_j)| f(\xi_i, \eta_j). \quad (5)$$

- Thus, the integral is evaluated numerically by a **double summation**, using the same weights and quadrature points as in one-dimensional quadrature. This involves two nested do loops.

2×2 Gauss Quadrature Used for Q4 Elements

Quadrature Point (ξ, η)	Weight (product of 1D weights)
$\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$1 \times 1 = 1$
$\left(+\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$1 \times 1 = 1$
$\left(+\frac{1}{\sqrt{3}}, +\frac{1}{\sqrt{3}}\right)$	$1 \times 1 = 1$
$\left(-\frac{1}{\sqrt{3}}, +\frac{1}{\sqrt{3}}\right)$	$1 \times 1 = 1$

 3×3 Gauss Quadrature Used for Q8 and Q9 Elements

Quadrature Point (ξ, η)	Weight (product)
$\left(-\sqrt{3/5}, -\sqrt{3/5}\right)$	$\frac{5}{9} \times \frac{5}{9} = \frac{25}{81}$
$\left(0, -\sqrt{3/5}\right)$	$\frac{8}{9} \times \frac{5}{9} = \frac{40}{81}$
$\left(+\sqrt{3/5}, -\sqrt{3/5}\right)$	$\frac{5}{9} \times \frac{5}{9} = \frac{25}{81}$
$\left(-\sqrt{3/5}, 0\right)$	$\frac{5}{9} \times \frac{8}{9} = \frac{40}{81}$

- Nodes:

- ① Node 1: $(0, 0)$,
- ② Node 2: $(1, 0)$,
- ③ Node 3: $(0, 1)$.

- Shape functions:

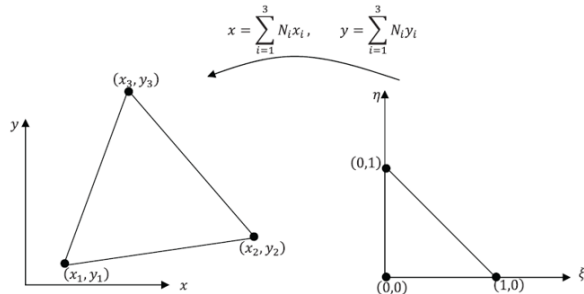
$$N_1 = 1 - \xi - \eta,$$

$$N_2 = \xi,$$

$$N_3 = \eta.$$

- The map from the parent element to the physical element is

$$x = \sum_{i=1}^3 N_i x_i, \quad y = \sum_{i=1}^3 N_i y_i.$$



Linear (T3) triangular element: physical domain (left) and parent domain (right).

- Nodes:

- Vertex nodes:

- 1 Node 1: $(\xi, \eta) = (0, 0)$,
 - 2 Node 2: $(\xi, \eta) = (1, 0)$,
 - 3 Node 3: $(\xi, \eta) = (0, 1)$.

- Mid-side nodes:

- 4 Node 4: midpoint of edge 1 $\leftrightarrow$ 2, $(\xi, \eta) = (\frac{1}{2}, 0)$,
 - 5 Node 5: midpoint of edge 2 $\leftrightarrow$ 3, $(\xi, \eta) = (\frac{1}{2}, \frac{1}{2})$,
 - 6 Node 6: midpoint of edge 3 $\leftrightarrow$ 1, $(\xi, \eta) = (0, \frac{1}{2})$.

- Shape functions:

$$N_1 = (1 - \xi - \eta)(1 - 2\xi - 2\eta),$$

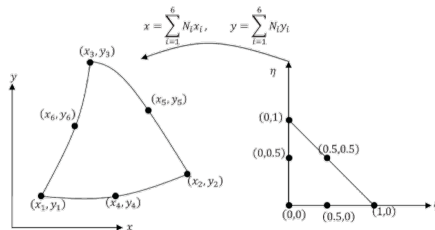
$$N_2 = \xi(2\xi - 1),$$

$$N_3 = \eta(2\eta - 1),$$

$$N_4 = 4\xi(1 - \xi - \eta),$$

$$N_5 = 4\xi\eta,$$

$$N_6 = 4\eta(1 - \xi - \eta).$$



Quadratic (T6) triangular element: physical domain (left) and parent domain (right).

Applications of T6

The T6 element is commonly used for problems requiring higher accuracy, such as:

- Structural analysis,
- Heat transfer,
- Fluid flow,
- Electromagnetic field simulations.

- For general curved-sided triangular elements, the numerical integration procedures are somewhat different than those for quadrilaterals. The integration formula is given by

$$I = \int_{\Omega^e} f \, d\Omega = \sum_{i=1}^{n_{gp}} W_i |\mathbf{J}^e(\xi_i)| f(\xi_i), \quad (8)$$

where the Jacobian is

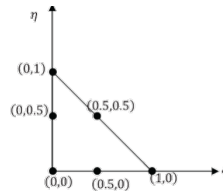
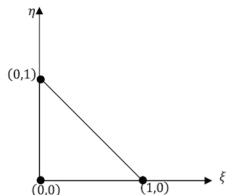
$$\mathbf{J}^e = \begin{bmatrix} \frac{\partial x}{\partial \xi_1} & \frac{\partial y}{\partial \xi_1} \\ \frac{\partial x}{\partial \xi_2} & \frac{\partial y}{\partial \xi_2} \end{bmatrix} = \begin{bmatrix} \sum_I^{n_{en}} \frac{\partial N_I^T}{\partial \xi_1} x_I^e & \sum_I^{n_{en}} \frac{\partial N_I^T}{\partial \xi_1} y_I^e \\ \sum_I^{n_{en}} \frac{\partial N_I^T}{\partial \xi_2} x_I^e & \sum_I^{n_{en}} \frac{\partial N_I^T}{\partial \xi_2} y_I^e \end{bmatrix}, \quad (9)$$

with the notation $\xi_1 \equiv \xi$, $\xi_2 \equiv \eta$.

- The weights and quadrature points for triangular elements are summarized in Table 7.7 on the following slide.

Table 7.7 Gauss quadrature weights and points for triangular domains.

Integration order	Degree of precision	ξ_1	ξ_2	Weights
Three-point	2	0.1666 666 666	0.1666 666 666	0.1666 666 666
		0.6666 666 666	0.1666 666 666	0.1666 666 666
		0.1666 666 666	0.6666 666 666	0.1666 666 666
Seven-point	5	0.1012 865 073	0.1012 865 073	0.0629 695 903
		0.7974 269 853	0.1012 865 073	0.0629 695 903
		0.1012 865 073	0.7974 269 853	0.0629 695 903
		0.4701 420 641	0.0597 158 717	0.0661 970 764
		0.4701 420 641	0.4701 420 641	0.0661 970 764
		0.0597 158 717	0.4701 420 641	0.0661 970 764
		0.3333 333 333	0.3333 333 333	0.1125



- The two basic categories of three-dimensional elements are **hexahedral** and **tetrahedral** elements. The former are generalisations of quadrilateral elements, whereas the latter are generalisations of triangular elements.
- Hexahedral elements:** The parent element domain of the eight-node hexahedral (or brick) element is a biunit cube with element coordinates ξ , η and ζ . The map to the physical domain is

$$\begin{aligned} x(\xi, \eta, \zeta) &= \mathbf{N}^{8H}(\xi, \eta, \zeta) \mathbf{x}^e, \\ y(\xi, \eta, \zeta) &= \mathbf{N}^{8H}(\xi, \eta, \zeta) \mathbf{y}^e, \\ z(\xi, \eta, \zeta) &= \mathbf{N}^{8H}(\xi, \eta, \zeta) \mathbf{z}^e, \end{aligned} \quad (10)$$

where $\mathbf{N}^{8H}(\xi, \eta, \zeta)$ are the eight-node hexahedral shape functions defined in the parent coordinate system shown in Figure 7.26.

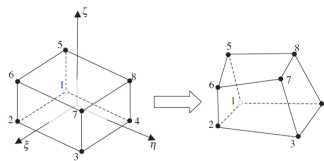


Figure 7.26 Mapping of the eight-node hexahedral from the parent to the physical Cartesian coordinate system. Nodes at $\zeta = -1$ are first numbered counterclockwise, followed by nodes at $\zeta = 1$.

Figure 7.26 Mapping of the eight-node hexahedral from the parent to the physical Cartesian coordinate system. Nodes at $\zeta = -1$ are first numbered counterclockwise, followed by nodes at $\zeta = 1$.

- The shape functions for an **eight-node hexahedral element** ($\mathbf{N}^{8H}(\xi, \eta, \zeta)$) in the natural coordinate system (ξ, η, ζ) are given by:

$$N_i(\xi, \eta, \zeta) = \frac{1}{8}(1 + \xi_i\xi)(1 + \eta_i\eta)(1 + \zeta_i\zeta) \quad (11)$$

for $i = 1, 2, \dots, 8$, where:

- (ξ, η, ζ) are the local coordinates in the element,
- (ξ_i, η_i, ζ_i) are the natural coordinate values of the i th node, which take values of ± 1 based on the node location.

Node Numbering and Corresponding Local Coordinates:

Node	ξ	η	ζ
1	-1	-1	-1
2	+1	-1	-1
3	+1	+1	-1
4	-1	+1	-1
5	-1	-1	+1
6	+1	-1	+1
7	+1	+1	+1
8	-1	+1	+1

- The Jacobian matrix $\mathbf{J}^e$ in three dimensions is

$$\mathbf{J}^e = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix}. \quad (12)$$

- The integral over a hexahedral element domain can be expressed as

$$\begin{aligned} I &= \int_{\Omega^e} f(\xi, \eta, \zeta) \, d\Omega = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 |\mathbf{J}^e(\xi, \eta, \zeta)| f(\xi, \eta, \zeta) \, d\xi \, d\eta \, d\zeta \\ &= \sum_{i=1}^{n_{\text{gp}}} \sum_{j=1}^{n_{\text{gp}}} \sum_{k=1}^{n_{\text{gp}}} W_i W_j W_k |\mathbf{J}^e(\xi_i, \eta_j, \zeta_k)| f(\xi_i, \eta_j, \zeta_k). \end{aligned} \quad (13)$$

$2 \times 2 \times 2$ Gauss Quadrature Rule for Hexahedral Elements

Gauss Point	ξ	η	ζ	Weight w
1	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	1.0
2	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	1.0
3	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	1.0
4	$-\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	1.0
5	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	1.0
6	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	1.0
7	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	1.0
8	$-\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	1.0

- The shape functions of the **four-node tetrahedral element** are given by

$$\begin{aligned} N_1^{4\text{Tet}} &= \xi_1, \\ N_2^{4\text{Tet}} &= \xi_2, \\ N_3^{4\text{Tet}} &= \xi_3, \\ N_4^{4\text{Tet}} &= \xi_4 = 1 - \xi_1 - \xi_2 - \xi_3. \end{aligned} \quad (14)$$

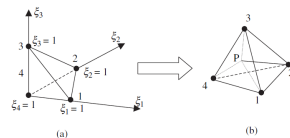


Figure 7.28 Mapping of the four-node tetrahedron from (a) the parent to (b) the physical Cartesian coordinate system. Also shown is the interior point P (not a node) in the physical domain (b).

Figure 7.28 Mapping of the four-node tetrahedron from (a) the parent to (b) the physical Cartesian coordinate system. Also shown is the interior point P (not a node) in the physical domain.

- Figure 7.29 shows the **10-node curved-face tetrahedral element**.
- Table 7.9 summarises the shape function construction for the ten-node tetrahedral element.

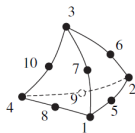


Figure 7.29 A 10-node curved-face tetrahedral element.

Figure 7.29 A 10-node curved-face tetrahedral element.

Table 7.9 Shape function construction for the 10-node tetrahedral element.

I	ξ_1	ξ_2	ξ_3	ξ_4	$N_I^{10\text{Tet}}$
1	1	0	0	0	$2\xi_1(\xi_1 - \frac{1}{2})$
2	0	1	0	0	$2\xi_2(\xi_2 - \frac{1}{2})$
3	0	0	1	0	$2\xi_3(\xi_3 - \frac{1}{2})$
4	0	0	0	1	$2\xi_4(\xi_4 - \frac{1}{2})$
5	$1/2$	$1/2$	0	0	$4\xi_1\xi_2$
6	0	$1/2$	$1/2$	0	$4\xi_2\xi_3$
7	$1/2$	0	$1/2$	0	$4\xi_1\xi_3$
8	$1/2$	0	0	$1/2$	$4\xi_1\xi_4$
9	0	$1/2$	0	$1/2$	$4\xi_2\xi_4$
10	0	0	$1/2$	$1/2$	$4\xi_3\xi_4$

Table 7.10 Gauss quadrature weights and points for tetrahedral domains.

Integration order	Degree of precision	ξ_1	ξ_2	ξ_3	Weights
One-point	2	0.25	0.25	0.25	1
Four-point	3	0.585 410 20	0.138 196 60	0.138 196 60	0.25
		0.138 196 60	0.585 410 20	0.138 196 60	0.25
		0.138 196 60	0.138 196 60	0.585 410 20	0.25
		0.138 196 60	0.138 196 60	0.138 196 60	0.25
Five-point	4	0.25	0.25	0.25	−0.8
		1/3	1/6	1/6	0.45
		1/6	1/3	1/6	0.45
		1/6	1/6	1/3	0.45
		1/6	1/6	1/6	0.45

Next: FEM for Multidimensional Scalar Field Problems

Thanks for your attention!